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# ShangJhe Li

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## Education

**National Taiwan University** *Sep 2023 - Present*

BS in Computer Science and Information Engineering (Junior)

- GPA: 4.05/4.3
- Coursework: System Programming, Computer Security, Machine Learning

## Experience

**Secure Systems Researcher** Taipei, Taiwan

*NTU CSIE* *July 2025 - Present*

- Read and analyze 10+ papers about system security, reliability, and fault tolerance.
- Currently working on novel exception handling for rust driver in the Linux kernel.

**Teaching Assistant (Data Structures and Algorithms)** Taipei, Taiwan

*NTU CSIE* *March 2025 - June 2025*

- Prepare the last programming problem for Homework 2. (Simulate bicycle parking slots as a tree)
- Help students with their programming assignments.

**Teaching Assistant (Introduction to Computer Programming)** Taipei, Taiwan

*NTU CSIE* *September 2025 - December 2025*

- Prepare three exams as well as three mock exams.

## Projects

**Owo Judge (Backend Developer)** Github Link

- Developed a containerized, multi-language backend for a competitive programming platform, featuring a decoupled architecture and MongoDB. The user can submit their code via git.
- Tools Used: TypeScript, Node.js, MongoDB, ioi/isolate, Docker
- Weakness: Currently, the system is only for DSA course. Some advanced features were not implemented.

**Challenge2025** Github Link

- Design the game for high school students to write python agents and learn how to collaborate with a medium-sized team (13 people).
- Tools Used: Godot, Git, Python
- Weakness: We use socket programming to implement the communication between the game server and the agents, which cause the performance and reliability issue.

**PseudoCode Interpreter** Github Link

- Implemented a pseudo-code interpreter with a recursive abstract syntax tree (AST) and polymorphism.
- Tools Used: C++, Object-Oriented Programming, Data Structures.
- Weakness: The use of string instead of enum for the type field causes the performance issue.

## Technical Skills

- Languages: C/C++, Python, TypeScript, Rust
- Technologies: Unix System Programming, Object-Oriented Programming, Data Structures, Algorithms